Basic Program Functions

Product: MASC Multi Axis Stepper Controller

Product Description

Stepper Motor Controller

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1 IMPORTANT SAFETY INFORMATION!

This document is intended as a user guide to operation as determined by the design.

Position control systems are inherently hazardous. Even a small motor, if coupled to a leadscrew, gearbox, or any other form of mechanism that provides a mechanical advantage, can generate considerable force and could cause serious injury. Incorrect operation can also lead to damage to the motor or associated machinery. It is essential that the purpose of the various fault detection features built into the MASC's ST5-Q controller be fully understood and used correctly.

As such, Installation must be carried out only by competent persons who are familiar with electronic control systems

2 MASC SETUP

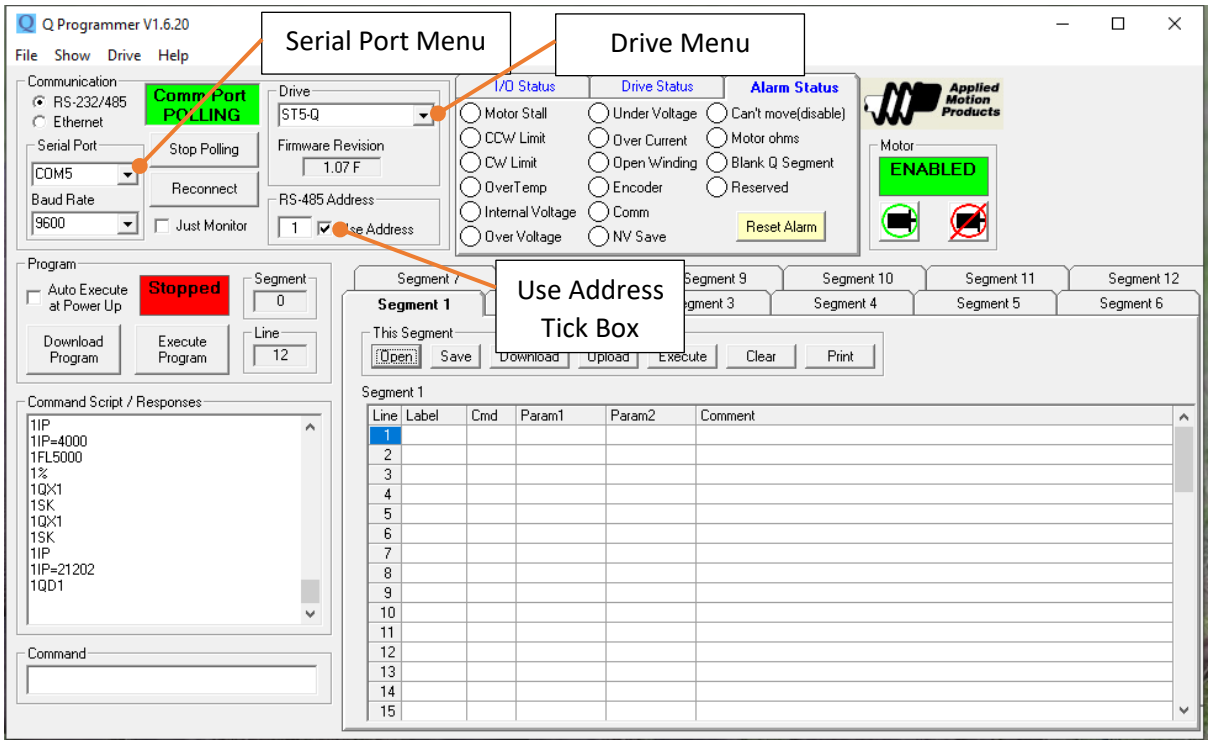
The MASC controller uses an Applied Motion Products ST5-Q controller to run the stepper motor. We recommend using the ST configurator and Q programmer software to setup and run the controller.

The software is free to download from www.applied-motion.com/products/software

The Applied Motion Products Q Programmer software gives the ability to write complex sequences and programs that can be stored and called up at specific times. For further information please see the Applied Motion Products manuals, available from www.applied-motion.com/support.

2.1 Launching Q Programmer

1. Open the software from the PC or laptop. ***Note: the software should always be launched before the MASC is switched on.***

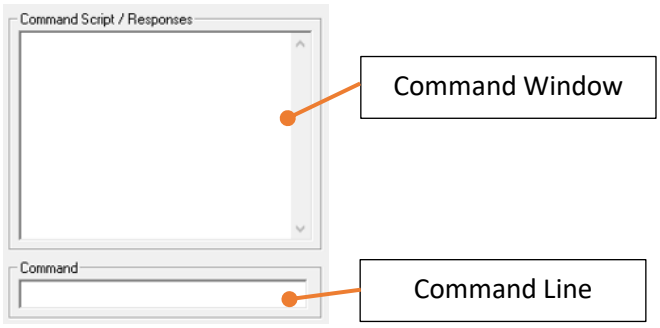


2. The software should detect a new Serial Port that is available. Use the drop down menu to select the correct Serial Port.
3. The Correct stepper driver type must be selected. Use the Drive Menu and select ST5Q.
4. Ensure that the Use Address check box is ticked.
5. Turn the MASC on.

2.2 Changing MASC Parameters

The Q programmer can be used to change the factory set values for many of the parameters, or perform move commands, as well as writing and executing programs and sequences. Any user wishing to make changes should fully read and understand the Applied Motion Products user manuals to avoid damage to the MASC or device.

All commands must be issued In Upper Case letters. Type commands into the Command line. Information returned from the MASC will be displayed in the Command Script/Return window.



To change a value, type the desired parameter, followed by the desired value and press return.

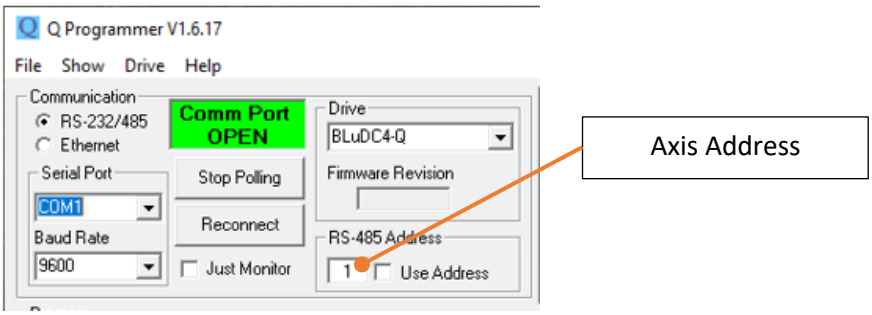
User Types	Drive Sends
VE5.5	-

To query a value, type the desired characters and press return. The software will report back with a value.

User Types	Drive Sends
VE	VE =5.5

2.3 Common Basic Parameters & Commands

If more than one axis system is used, ensure that the axis address is set to the corresponding MASC before issuing commands.



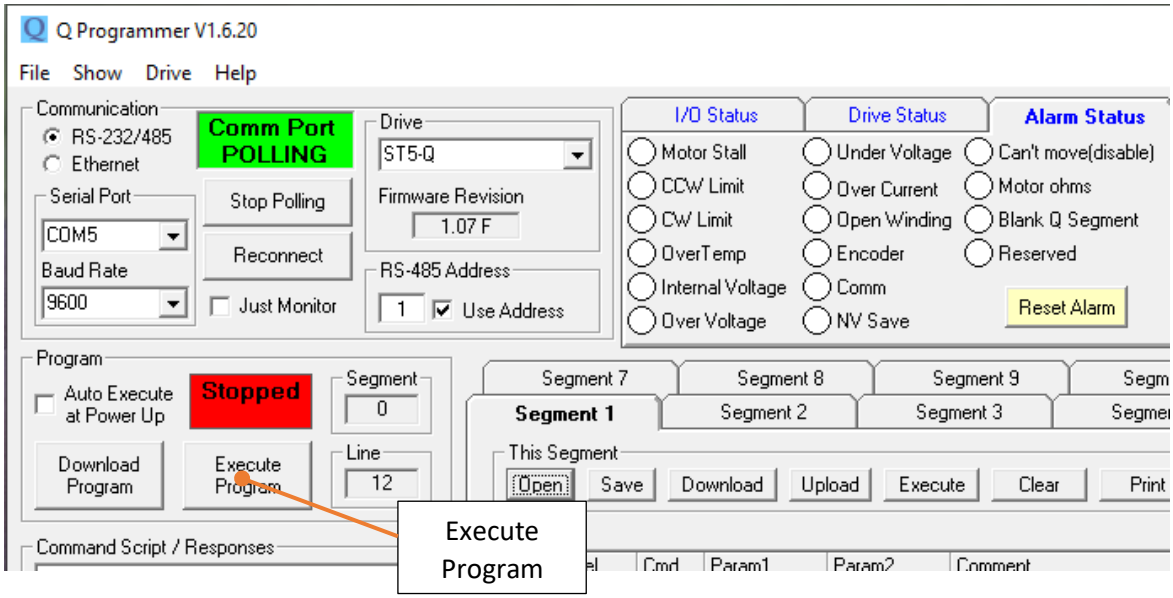
Command	Parameter
VE	Motor Shaft velocity (revs/sec)
AC	Acceleration rate (revs/sec2)
DE	Deceleration rate (revs/sec2)
CC	Change current used during moves (Amps)
CI	Change Idle current (Amps)
FL	Move relative to current position (counts)
FP	Move to absolute position
IP	Query position
EG	Electronic Gearing (Equivalent to Microstep Resolution).
SA	Save parameters
ST	Stop current command
SH3R	Seek Home
SP0	Set Position to Zero

3 Linear Encoder Function

If the MASC has been purchased with a UHV design product equipped with a linear encoder, then the standard stall prevention/detection encoder functions of the MASC drive do not give precise enough control. In such a case the MASC comes pre-loaded with a program to enable use of a linear encoder with error correction function.

3.1 Operation

1. Launch Q programmer software.
2. Press execute program button



3. The program will now be running
4. To home the device type RL2987654321 into the command line and press enter (device will drive to the limit switch and then to the home switch, allow the cycle to finish). The home position will be set to 0
5. To perform relative moves use RL2xxxxxx Values are in encoder counts (see calculator for counts/mm)
6. To perform absolute moves use RL1xxxxxx Values are in encoder counts (see calculator for counts/mm)

3.2 Adjusting settings

Command	Parameter	Multiplier (
RLV	Motor Shaft velocity (revs/sec)	240 (2 revs/sec = 240 x 2 = 480)
RLA	Acceleration rate (revs/sec/sec)	6 (100 revs/sec/sec = 6 x 100 = 600)
RLB	Deceleration rate (revs/sec/sec)	6 (100 revs/sec/sec = 6 x 100 = 600)
RL1	Input command position (Absolute)	N/A
RL2	Input command position (Relative)	N/A
RL2987654321	Seek home	
RL3	Input tolerance window (encoder counts)	N/A
RU51	Query drive status: 0 = Idle 1 = Moving 2= Seeking home 3 = Correcting 4 = Not reached command position	N/A
RL6	Maximum number of correction attempts	N/A
IE	Query encoder position	N/A

The above commands will only work whilst the program is still running. To stop the program, Press the stop program button. Commands may then be entered as described in section 2.3

3.3 Checking

The encoder movement should be checked against the expected values. To do this follow the steps below.

1. Drive to home (or a known position.
2. Measure the device with a vernier or rule to establish a “zero” position.
3. Use the commands above to request a position move of 10mm
4. Measure with a vernier or rule again to check that a 10mm move has actually been performed.
5. If position is not achieved, check encoder settings and repeat.

3.3.1 Troubleshooting

The most common problem will be that the motor will be driving the wrong way with respect to the encoder. This normally presents as the drive making smaller and smaller moves in the same direction and then when measuring it’s position, it has not moved the desired amount.

1. Press Stop Program
2. Type EP0 and press return to zero the encoder position
3. Type SP0 and press return to zero the motor position
4. Type FL10000 and press return to move the motor.
5. Type IP and press return. The motor position should be the 10000 counts commaned
6. Type IE and press return. The encoder positon value will not match but the important thing is that the encoder does no rea a minus value. If this is the case, then go to the next line.
7. Type ED into the command line and press return.
8. ED0 is the default direction
9. Type ED1 and press return if the reverse direction if required.
10. Type SA and press return to save the setting change
11. Repeat steps 1 -6 to check.

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